

Danzel Yap

Project Engineer — LA Convention Center

Diamond Bar, CA 91765 | (408) 335-3615 | dyap123@gmail.com

SUMMARY

Project Engineer on the Los Angeles Convention Center (LACC) expansion, managing structural foundation pours from layout through placement: RFIs, submittals, quantity takeoffs, budget tracking, and look-ahead scheduling. Built and deployed a suite of AI-powered construction management tools adopted by the project team, streamlining processes and saving the team hundreds of hours per week.

SKILLS

Project Engineering: RFIs, submittals, quantity takeoffs, concrete takeoffs, pour planning and logistics, embed coordination, budget and cost tracking, look-ahead scheduling, subcontractor coordination, MEP coordination, drawing review, field quality control, daily reporting, Revit.

Construction Software: Bluebeam Revu, Microsoft Excel, Microsoft Office, Google Workspace, Autodesk Revit.

Programming and AI: Python, JavaScript, C#, Java, MATLAB, VBA, LLM API integration (Anthropic, OpenAI, Gemini), Firebase, React, Google Apps Script

Design: SolidWorks, Autodesk Fusion 360. Autodesk Revit.

Languages: English (fluent), Mandarin (fluent), Cantonese (conversational)

EXPERIENCE

Webcor Craft

Project Engineer, Los Angeles, CA | [Month] 2025 - Present

- Manage structural foundation pours on the \$2B Los Angeles Convention Center (LACC) expansion, coordinating concrete operations, embed installation, quantity takeoffs, and pour logistics from layout through placement.
- Process RFIs and submittals, perform concrete quantity takeoffs, track budgets, and maintain look-ahead schedules across concurrent pour sequences.
- Coordinate installation and verification of approximately 300 embeds across 75 embed types, tracked through a custom plan-map application built in-house.
- Designed, built, and deployed a suite of 6 AI-powered construction management applications adopted by the project management team, saving each project engineer approximately one day per week (detailed under Projects).
- Led evaluation of a third-party AI software vendor and recommended against adoption, saving the company \$75,000.

Clune Construction Company

Project Management Intern, San Francisco, CA | Jun 2024 - Aug 2024

- Analyzed construction drawings, basic MEP coordination, developed project budgets and schedules, and researched practical applications of AI in construction project management.

Cal Poly Pomona Formula E

Nuts and Bolts Lead (School Project), Pomona, CA | Sep 2022 - Jun 2024

- Led design and manufacture of mechanical systems (chassis, brakes, driver ergonomics) for a competition electric vehicle.

K-12 STEM Center USA

STEM Instructor (Part-Time), Claremont, CA | Oct 2022 - Mar 2026

- Taught robotics, Arduino, programming, and soldering to K-12 students; developed and launched an introductory AI course.

PROJECTS

AI in Construction Suite | 2024 - Present

- Built and deployed 6 web applications used daily by a live project team (pour tracking, embed tracking, tool inventory, crew attendance, scheduling, cylinder break tracking), plus an AI chat assistant integrated with Google Workspace, Discord, and Microsoft Teams.

FIRST Robotics Competition | California | Nov 2018 - Jun 2021

- Designed and built competition robots on a multidisciplinary team; secured team funding and delivered builds to competition deadlines.

EDUCATION

Bachelor of Science, Mechanical Engineering — California State Polytechnic University, Pomona | December 2025